

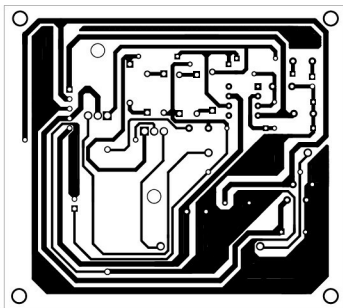


Fig. 5: Actual-size PCB layout of the 30W audio power amplifier

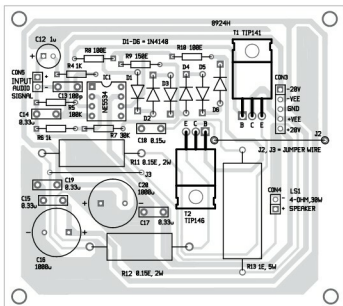


Fig. 6: Components layout for the PCB in Fig. 5

ZD4 to provide $\pm 20V$ for op-amp NE5534. Though the maximal power supply for NE5534 is $\pm 22V$, it should not go beyond $\pm 20V$.

The circuit does not need any adjustment to work.

Construction and testing

An actual-size PCB layout of the power supply circuit is shown in Fig. 3 and its components layout in Fig. 4. Actual-size PCB layout of the 30W

audio power amplifier using NE5534 and Darlington transistors is shown in Fig. 5 and its components layout in Fig. 6.

Assemble the circuits on their respective PCBs. Connect primary terminals (L and N) of the transformer to L (line) and N (neutral) marked on the power supply PCB at X1. Similarly, connect secondary terminals of X1 to 22V, 0, 22V marked on the PCB.

On the amplifier PCB, connect CON2 (from the power-supply PCB) to CON3 using suitable 5-core wires for power connections. Connect a 30W, 4-ohm speaker to CON4. At CON5, feed audio signal from any audio source like mobile phone.

Now, connect 230V AC mains power supply to CON1. When you switch on the circuit through switch S1, you should be able to hear amplified sound from the speaker connected at CON4. **EFY**

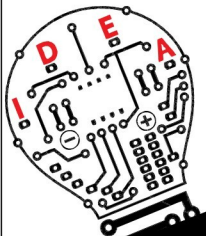
Petre Tzvet Petrov was a researcher and assistant professor in Technical University of Sofia (Bulgaria) and expert-lecturer in OFFPPT (Casablanca), Kingdom of Morocco. Now he is working as an electronics engineer in the private sector in Bulgaria.



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